

LZW Compression Challenge

Input Sequence (Length 12):

A	A	A	A	A	A	C	A	A	A	A	A
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Task Requirements:

1. Trace the LZW algorithm.
2. Initial dictionary: A(0), C(1), T(2), G(3).
3. Show all dictionary insertions.
4. Compare ASCII vs LZW using 4-bit codes.

Examiner Solution Key

Current w	Next c	New Entry	Index	Output
A	A	AA	4	0
AA	A	AAA	5	4
AAA	C	AAAC	6	5
C	A	CA	7	1
AAA	A	AAAA	8	5
AA	EOF	-	-	4

Final Statistics:

Dictionary Size: 9

Encoded Output: [0, 4, 5, 1, 5, 4]

Original Size: 96 bits

Compressed Size: 24 bits

Compression Savings: 75.0%